3. How the exchange rate is determined: demand and supply

3.1 Finding the equilibrium exchange rate

In Section 2, we saw that people and businesses demand US dollars to make payments abroad, while dollars enter the Indian foreign exchange market through sources such as export earnings and remittances. How do these demands and supplies determine the exchange rate?

We can answer this question using a demand-and-supply model. The demand curve shows the quantity of dollars that buyers wish to purchase at different exchange rates. The supply curve shows the quantity of dollars that sellers wish to supply at those rates.

The two curves meet at a point called the equilibrium. At this exchange rate, the quantity of dollars demanded equals the quantity supplied. There is no excess demand or excess supply at the intersection, so the model identifies no pressure for the exchange rate to change from that point.

To represent this in a diagram, put the exchange rate, measured as rupees per US dollar (₹/$), on the vertical axis and the quantity of dollars traded on the horizontal axis. The downward-sloping demand curve and upward-sloping supply curve intersect at the equilibrium exchange rate. The position of this intersection depends on both curves, not on either curve alone.

3.2 What happens when demand or supply changes?

The equilibrium exchange rate need not remain at the same level. When the demand for dollars or their supply changes, the curves shift, and the market moves towards a new equilibrium.

An increase in the demand for dollars

Suppose more Indians begin travelling to the United States or purchasing foreign goods and services. They need more dollars to make these payments. At each exchange rate, buyers now wish to purchase more dollars than before. The demand curve therefore shifts to the right.

Consider a hypothetical market in which the initial equilibrium exchange rate is ₹82/$. After the increase in demand, suppose the new equilibrium is ₹86/$. Buyers must now pay ₹86 rather than ₹82 for each dollar. The price of the dollar in rupees has risen, which means that the rupee has depreciated against the dollar.

This is the result shown in the second panel of Figure 3.1. The supply curve remains unchanged, while the demand curve shifts to the right. The new intersection lies at a higher exchange rate.

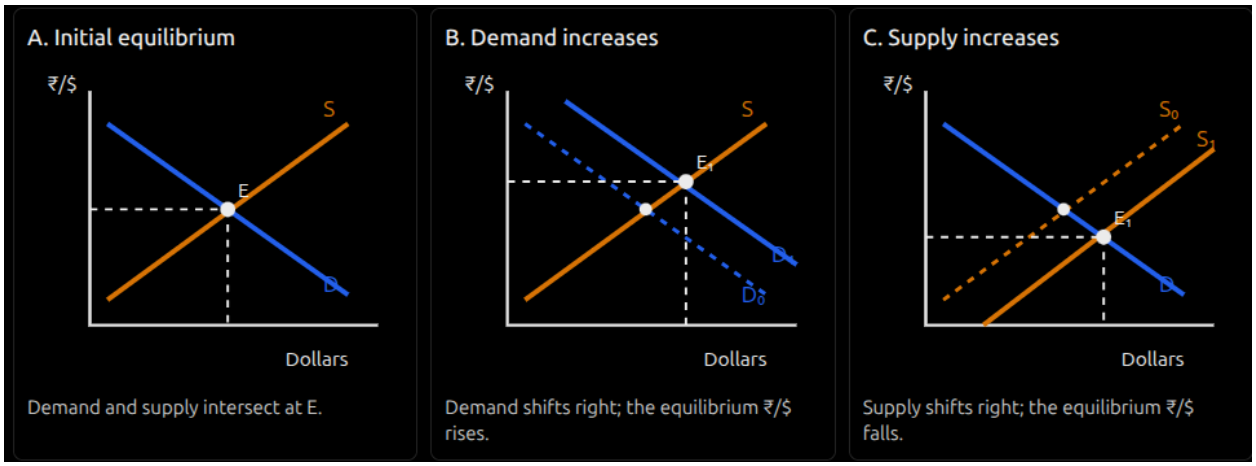
An increase in the supply of dollars

Now consider the opposite situation. Suppose Indian exporters receive more dollar payments from customers abroad, increasing the supply of dollars in the Indian foreign exchange market. At each exchange rate, sellers are willing to supply more dollars than before, so the supply curve shifts to the right.

Suppose the initial equilibrium is ₹82/$ and the increased supply brings the exchange rate down to ₹79/$. Each dollar can now be exchanged for fewer rupees. The rupee has appreciated against the dollar.

The third panel of Figure 3.1 illustrates this change. The demand curve remains unchanged, while the supply curve shifts to the right. Its intersection with the demand curve now occurs at a lower exchange rate.

Figure 3.1 - How changes in demand and supply affect the exchange rate



3.3 Using the model to understand exchange-rate movements

These examples isolate one change at a time. In the first, demand increases while supply is held constant; in the second, supply increases while demand is held constant. This makes it possible to identify how each change affects the equilibrium exchange rate.

Economists describe this method as assuming that other things remain constant. It allows us to examine the effect of one change without confusing it with other changes occurring at the same time.

Actual foreign exchange markets are more complicated. Demand and supply may shift together. For example, stronger demand for imports may increase the demand for dollars, while higher export earnings increase their supply. The resulting exchange rate depends on the combined effects of these changes. As the demand-and-supply model shows, we must consider both curves to understand where the new equilibrium will lie.

4. Appreciation and depreciation: who wins, who loses?

A change in the exchange rate does not affect everyone in the same way. The result depends on whether a person or business needs to buy foreign currency to make payments or receives foreign currency as income.

4.1 When the rupee depreciates

Suppose the exchange rate rises from ₹82/$ to ₹86/$. The rupee has depreciated because each US dollar now costs more rupees. This changes the cost of international transactions.

An Indian importer who pays a supplier in dollars must spend more rupees for the same dollar payment. The same applies to Meera, who is preparing to study in the United States: if her college fees are fixed in dollars, their rupee cost rises when the rupee depreciates.

For an Indian exporter, however, depreciation may be advantageous. Selvi runs a garment business in Tiruppur and receives payments from foreign buyers in dollars. When those dollars are converted into rupees, each dollar brings in more rupees. Indian goods may also become cheaper for foreign buyers when their prices are expressed in dollars, improving exporters' price competitiveness. The size of this advantage depends on how buyers respond to price changes.

Ravi, whose uncle works in Dubai and sends money to India, illustrates another effect. If the remittance is sent in a foreign currency and its value against the rupee rises, the same amount of foreign currency converts into more rupees for the recipient.

4.2 When the rupee appreciates

If the exchange rate falls from ₹86/$ to ₹82/$, the rupee appreciates. The effects are broadly reversed. Importers and students paying dollar-denominated expenses need fewer rupees to buy the same number of dollars. Meera's fees, for example, become less expensive in rupee terms.

Selvi may face the opposite effect: her dollar earnings convert into fewer rupees, and her garments may become more expensive for foreign buyers if their rupee prices remain unchanged. Ravi's family may likewise receive fewer rupees for the same foreign-currency remittance.

4.3 Why the effects differ

The table summarises the main effects.

Group	Rupee depreciation	Rupee appreciation
Importers	Imports cost more in rupees	Imports cost less in rupees
Students paying in dollars	Expenses rise in rupee terms	Expenses fall in rupee terms
Indian exporters	May gain price competitiveness	May lose price competitiveness
Recipients of foreign-currency remittances	Receive more rupees for the same amount	Receive fewer rupees for the same amount

Neither appreciation nor depreciation benefits everyone. A weaker rupee can help an exporter earning dollars while increasing the costs of an importer spending dollars. A stronger rupee can reduce the cost of foreign purchases while lowering the rupee value of foreign-currency earnings. The effect of an exchange-rate movement therefore depends on a person’s or business’s position in international transactions.